Objective: The objective of this lab is to design RAM using combinational logic circuits. By having memory indexes and saving our inputs into the indices of our memory array using Vivados Verilog code and implementing the program onto the BASYS 3 circuit board to test our design.

Equipment: Vivado 2017-4 software used to program a BASYS 3 circuit board.

Pre-Lab (Truth Table):

A3	A2	A1	A0	O3/carry	O2/ sum1	O1 sum0	O0 (F1)
0	0	0	0	0	0	0	0
0	0	0	1	0	0	1	0
0	0	1	0	0	1	0	0
0	0	1	1	0	1	1	0
0	1	0	0	0	0	1	0
0	1	0	1	0	1	0	0
0	1	1	0	0	1	1	0
0	1	1	1	1	0	0	0
1	0	0	0	0	0	1	0
1	0	0	1	0	1	1	1
1	0	1	0	1	0	0	0
1	0	1	1	1	0	1	0
1	1	0	0	0	1	1	0
1	1	0	1	1	0	0	0
1	1	1	0	1	0	1	0
1	1	1	1	1	1	0	1

Procedure/Design: This program will take in 4 bits as data and a 4 bit address, 8 switch inputs and 1 save button input, 9 total inputs. Since the address is 4 bits, there are 2^4=16 available memory indices. In order to save the data the WE button must be pressed. If the data is saved into an address, it will show the output on LEDs of the data representation to the circuit board.

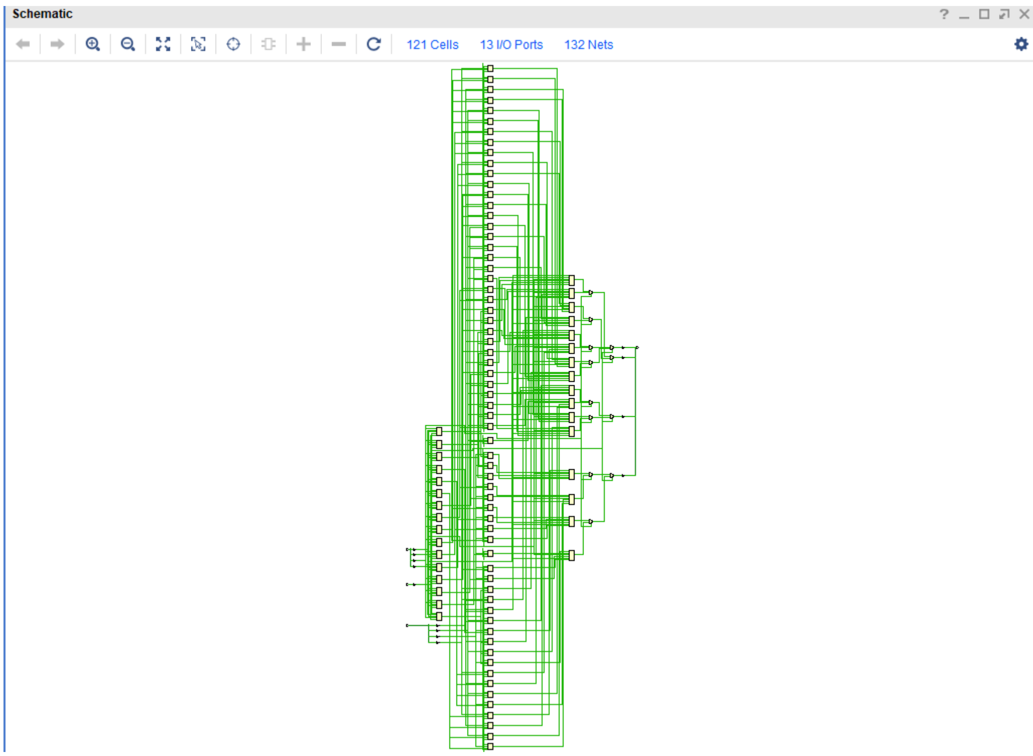
Source Code:

ram.v

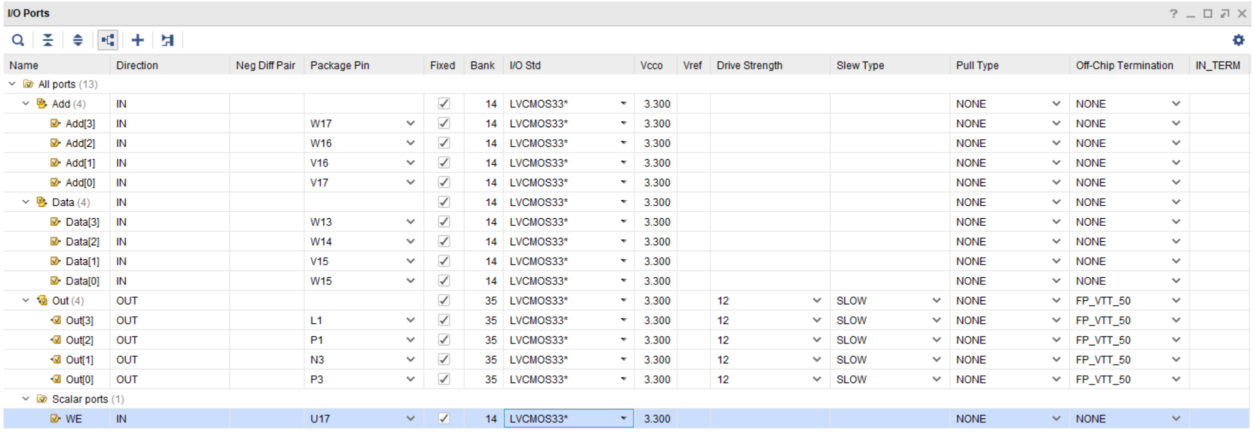
C:/Users/ry266542/Desktop/Lab/exp6/exp6.srcs/sources_1/new/ram.v

```
`timescale 1ns / 1ps


module ram(
input [3:0] Data,
input [3:0] Add,
output [3:0] Out,
input WE
);
reg [3:0] Mem [0:15];

always@(Data,Add)
begin
if(WE)
Mem[Add] <= Data;
end
assign Out = Mem[Add];
endmodule

```

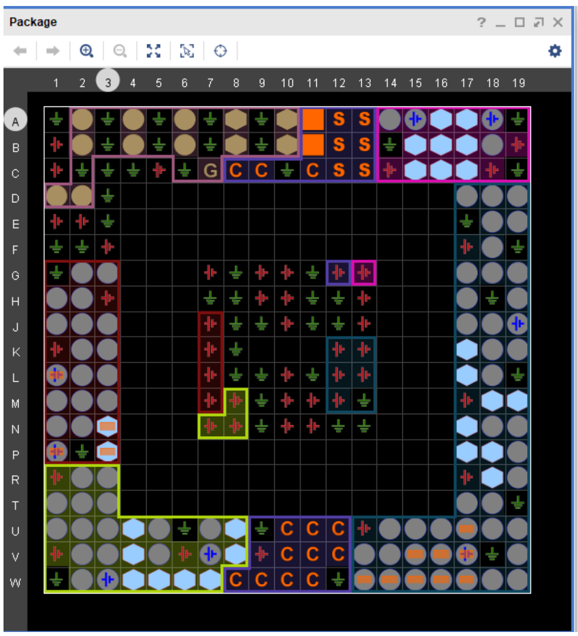
Schematic:



Assigned input/output pins:

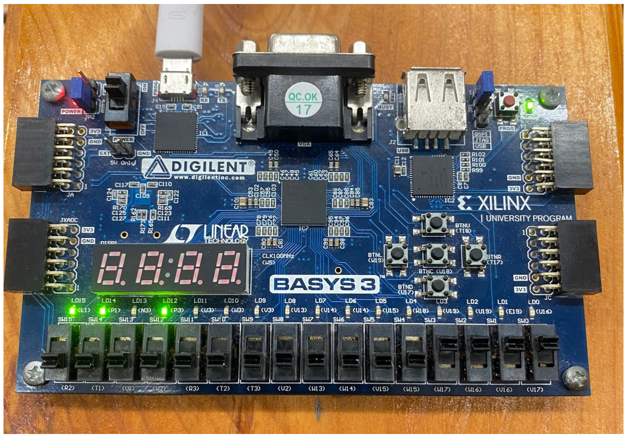

I/O Ports

Name	Direction	Neg Diff Pair	Package Pin	Fixed	Bank	I/O Std	Vcco	Vref	Drive Strength	Slew Type	Pull Type	Off-Chip Termination	IN_TERM
All ports (13)													
Add (4)	IN			✓	14	LVCMOS33*	3.300				NONE	NONE	
Add[3]	IN		W17	✓	14	LVCMOS33*	3.300				NONE	NONE	
Add[2]	IN		W16	✓	14	LVCMOS33*	3.300				NONE	NONE	
Add[1]	IN		V16	✓	14	LVCMOS33*	3.300				NONE	NONE	
Add[0]	IN		V17	✓	14	LVCMOS33*	3.300				NONE	NONE	
Data (4)	IN			✓	14	LVCMOS33*	3.300				NONE	NONE	
Data[3]	IN		W13	✓	14	LVCMOS33*	3.300				NONE	NONE	
Data[2]	IN		W14	✓	14	LVCMOS33*	3.300				NONE	NONE	
Data[1]	IN		V15	✓	14	LVCMOS33*	3.300				NONE	NONE	
Data[0]	IN		W15	✓	14	LVCMOS33*	3.300				NONE	NONE	
Out (4)	OUT			✓	35	LVCMOS33*	3.300		12	SLOW	NONE	FP_VTT_50	
Out[3]	OUT		L1	✓	35	LVCMOS33*	3.300		12	SLOW	NONE	FP_VTT_50	
Out[2]	OUT		P1	✓	35	LVCMOS33*	3.300		12	SLOW	NONE	FP_VTT_50	
Out[1]	OUT		N3	✓	35	LVCMOS33*	3.300		12	SLOW	NONE	FP_VTT_50	
Out[0]	OUT		P3	✓	35	LVCMOS33*	3.300		12	SLOW	NONE	FP_VTT_50	
Scalar ports (1)													
WE	IN		U17	✓	14	LVCMOS33*	3.300				NONE	NONE	

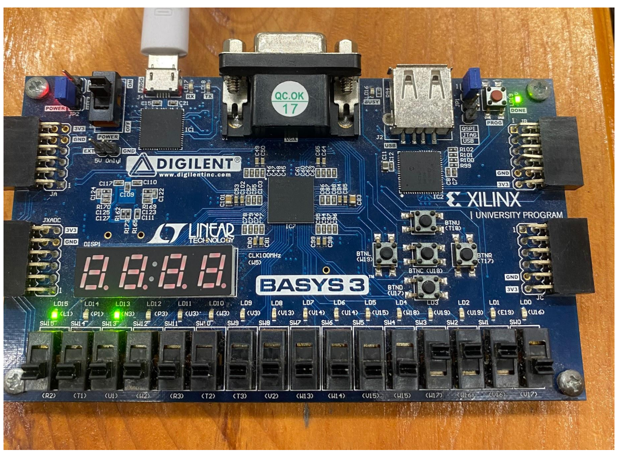


Results: The experiment lab matched the intended outcome of the truth table made in the Pre-lab, based on the two functions. Below are 4 different outputs on the circuit board matching our truth table.

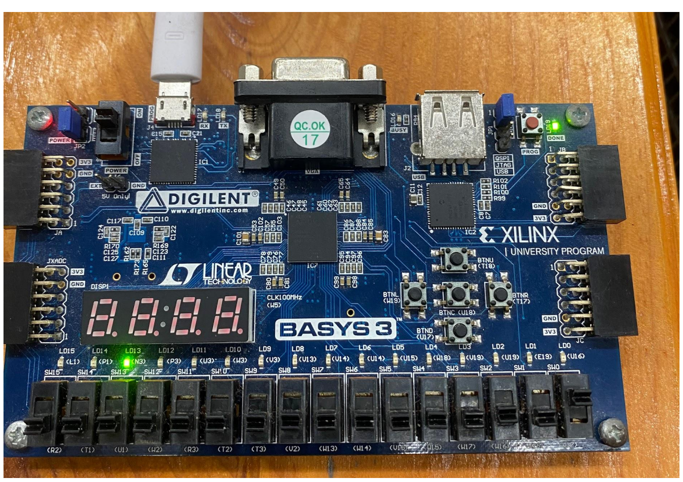
Memory Address: 1111
Data:1101



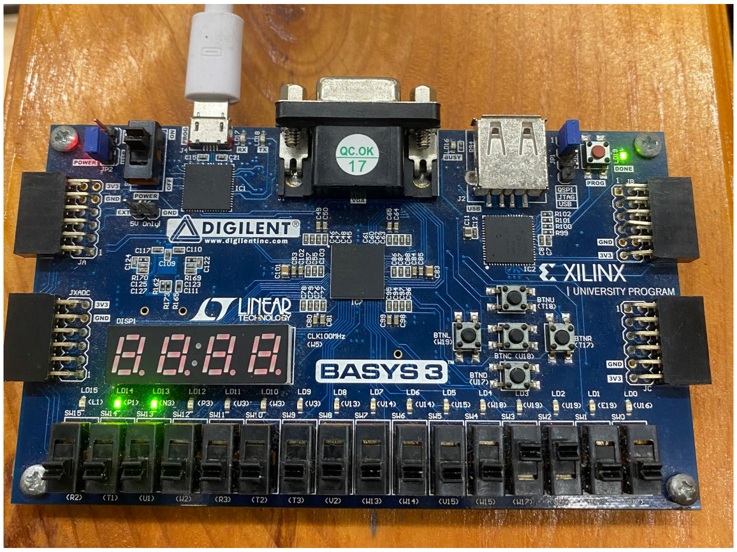
Memory Address: 1110
Data:1010



Memory Address: 0001
Data:0010



Memory Address: 1100
Data:0110



Conclusion: The experiment was successful giving the desired outputs. An advantage of RAM is that the data stored can be changed by reading and writing. ROM is permanent and cannot be changed. RAM can be changed but will be erased when turned off, so depending on particular situations one will be better over the other. All requirements of this lab have been met and have been tested by programming our circuit board. RAM/ROM is cost effective and improves efficiency by having memory saved. By having two 32x4 devices you can connect it by an enable wire 0 or 1 to create a 64x4 system to access the memory from the inputs.